

FCC 3101

CHAPTER 4

PART 2

4.4 STRUCTURAL DESIGN

4.4.1 Terms Used

4.4.1.1 Structural behavior refers to the three desirable characteristics of a structure: -

- a) **Stability:** The absence of any condition that may cause **premature failure**.
- b) **Safety:** The ability to sustain ultimate loads without **collapse**.
- c) **Serviceability:** The ability to remain **serviceable** under normal loads.

4.4.1.2 **Structural analysis** refers to the techniques used to predict the **structural response** due to a given **load scenario**. [The **worst load scenario** is called the **ultimate load scenario** and is used in structural design].

4.4.1.3 **Structural design** refers to all the engineering work which results in **creating on paper how the proposed structure must be constructed** so that its structural behavior will be **stable, safe and serviceable throughout its design life**.

4.4.2 Basis, Aim and Motto

4.4.2.1 The basis of structural design lies on **three premises**: -

1. All **stability conditions** must be satisfied.
[This premise ensures that the structure will be **stable**].
2. The structural response **must always be less** than the structural resistance by a **recommended factor of safety**.
[This premise ensures that the structure will be **safe**].
3. The **structural behavior** under normal loads must satisfy standard criteria for each serviceability condition.
[This premise ensures that the structure will be **serviceable**].

4.4.2.2 The aim of structural design is to achieve **acceptable probability** [usually 95% confidence level] that the designed structure will not become “unfit” for its **intended purpose** during its **design life**. [“Unfit” means either **unstable** or **unsafe** or **unserviceable** or any combination thereof].

4.4.2.3 The **motto** adopted in structural design is: **“Design for the worst load scenario and hope for the best structural behavior”**.

4.4.3 Requirements

Structural design must satisfy **four requirements**: -

4.4.3.1 Functional requirements: The structure must satisfy three functional requirements: -

- a) Spatial function:** The spaces enclosed by the structure must meet client requirements **[Arch]**
- b) Stability, safety and serviceability function:** The structure must be stable, safe and serviceable. **[SE]**
- c) Adaptability function:** The structure must be adaptable to other possible purposes. **[Arch and SE]**

- 4.4.3.2 Constructional requirements:** Constructional requirements relate to the financial and constructability constraints, namely: -
- a) Cost:** The initial cost and maintenance costs must be optimized **[Arch and SE]**
 - b) Time:** The construction process must fall on the most critical path **[CM]**
 - c) Quality:** The finished structure must meet laid down standards **[Arch and SE]**
 - d) Opportunity cost:** The structure must give the client maximum benefits at minimum costs. **[ALL]**

4.4.3.3 Aesthetic requirements: Aesthetic requirements relate to the esteem value of the finished structure. The structure **must marry (or be married)** into the existing environment in a manner that enhances **aesthetic appeal, esteem value and beauty. [Arch]**

4.4.3.4 Environmental requirements: Environmental requirements relate to the manner in which the structure will affect the existing environment. The structure, or its construction process, **must NOT degrade the existing environment** or adversely affect any of the beneficial eco-systems in the environment. **[Arch and SE]**

4.4.4 Design Process

Structural design is accomplished in **eight steps**: -

4.4.4.1 Appointment: The engineer is appointed by any of the following means: -

- a) Direct appointment by the client [individual clients and small projects].
- b) Competitive closed bidding [individual clients and medium projects].
- c) Expression of Interest, listing then appointment [corporate entities].
- d) Competitive open bidding [government agencies].

4.4.4.2 Client's brief: The engineer is given **working instructions** by the client. The working instructions (**are called client's brief**) and may be: -

- a) Verbal or written in layman's language.
- b) Written in technical language.
- c) The architectural drawings.

4.4.4.3 Site investigations: Site investigations are carried out by the engineer in order to establish: -

- a) The terrain and drainage pattern.
- b) Sub-soil properties and safe bearing capacity.
- c) Accessibility and availability of services.

[The engineer can submit **Fee Note No. 1**]

- 4.4.4.4 Structural modelling:** The engineer formulates several viable structural models for **appraisal**. Each model is appraised on the basis of its **technical** and **economic viability**. The **most viable model** (called the **optimum model**) is adopted.
- 4.4.4.5 Structural analysis and design:** The engineer evaluates the loads which will act on the structure. Structural analysis is carried out to **predict the structural response** of each member. The **members** and **their joints** are then specified in such a manner that the structural resistance **will always be greater than** the structural response.

- 4.4.4.6 **Structural drawings:** The engineer prepares detailed **structural drawings** showing member sizes, member details, joint details, material specifications, construction technology and all other relevant information.
- 4.4.4.7 **Approval:** The engineer submits the structural drawings, together with the design calculation, and all other requisite documents to the **construction regulatory bodies** for approval.
- 4.4.4.8 **Tendering:** The QS (assisted by Arch and SE) prepares **tender documents**. The tender documents are used to solicit for bids from eligible contractors. The bids are evaluated and the contract is awarded to the successful bidder. [The engineer can submit **Fee Note No. 2**]

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END OF PART 2